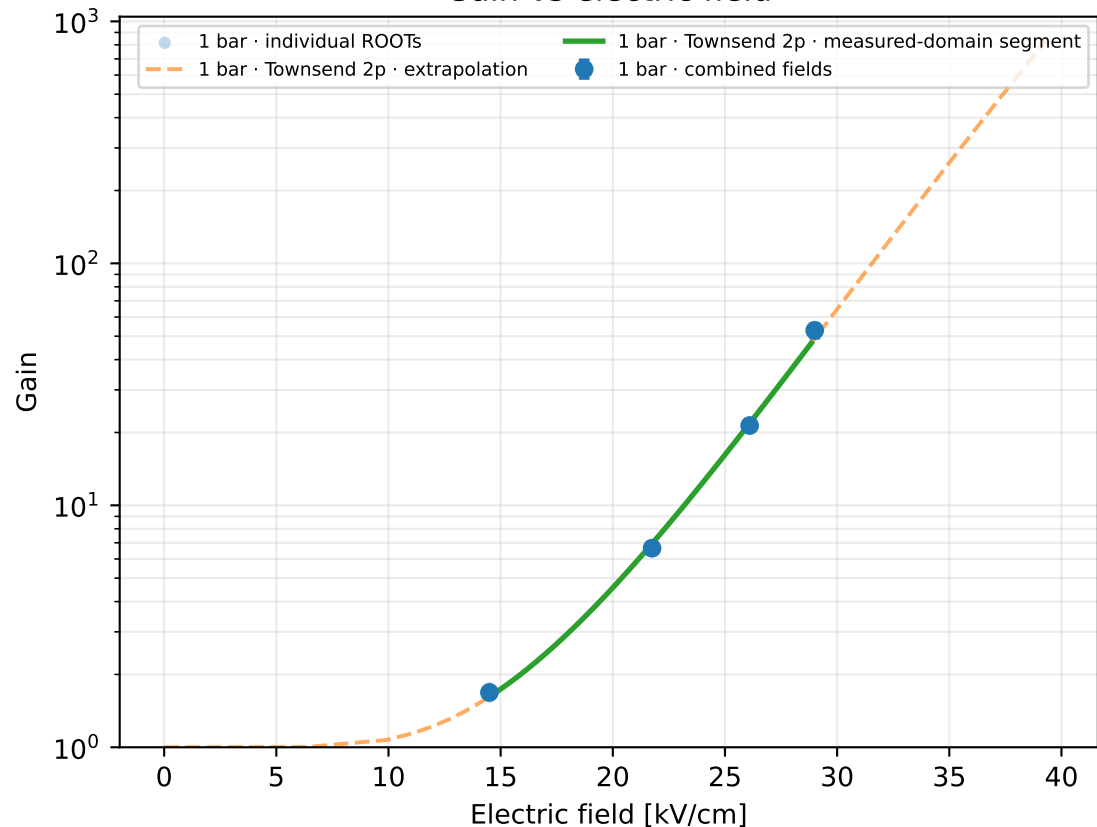
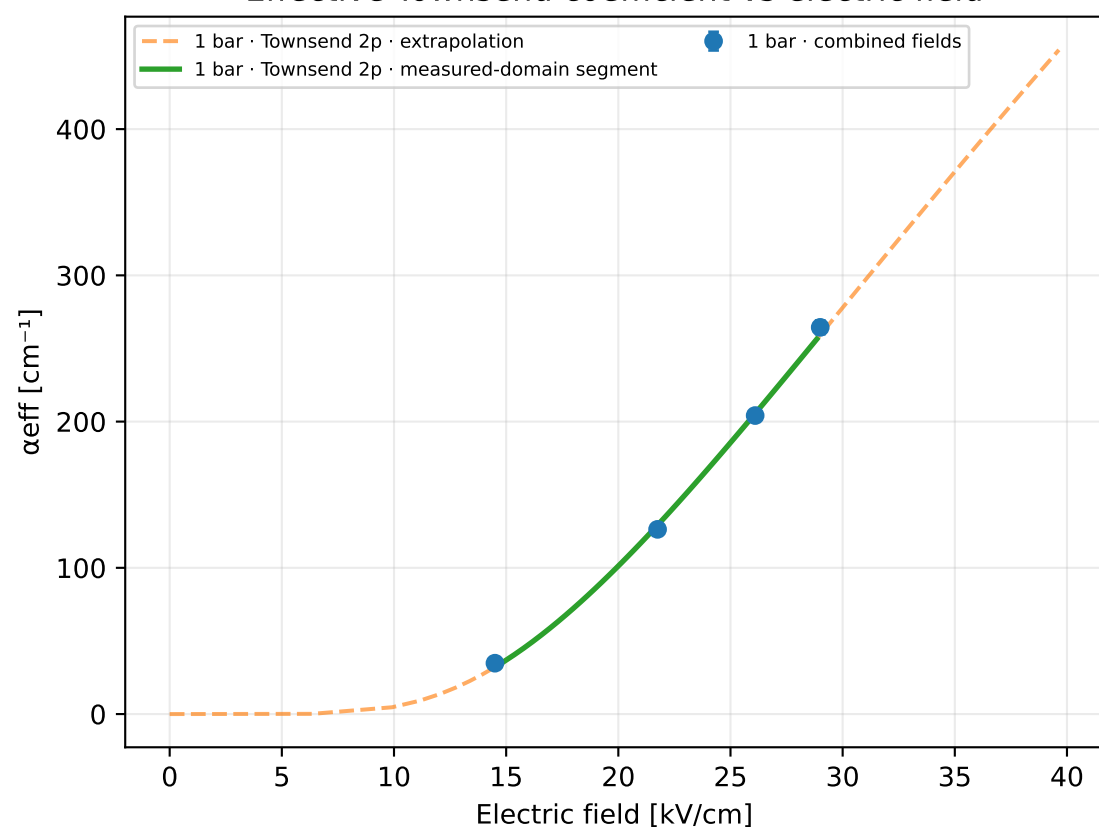


ArCF4Iso · Ar 95 % / CF₄ 3 % / Iso 2 % · gap 0.15 mm

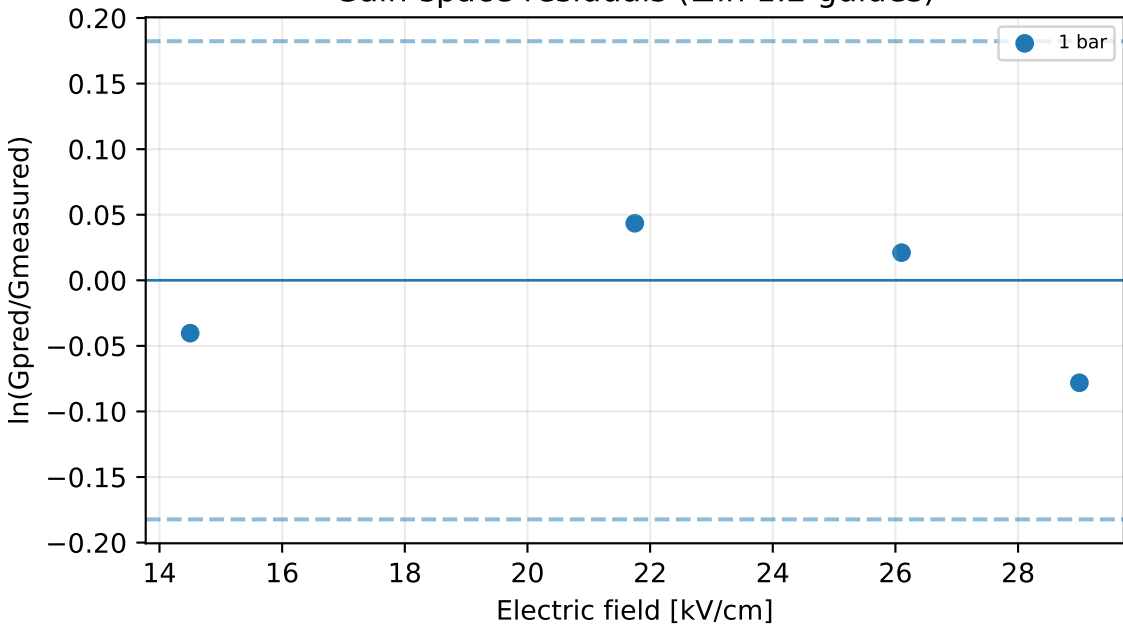
Gain vs electric field



Effective Townsend coefficient vs electric field



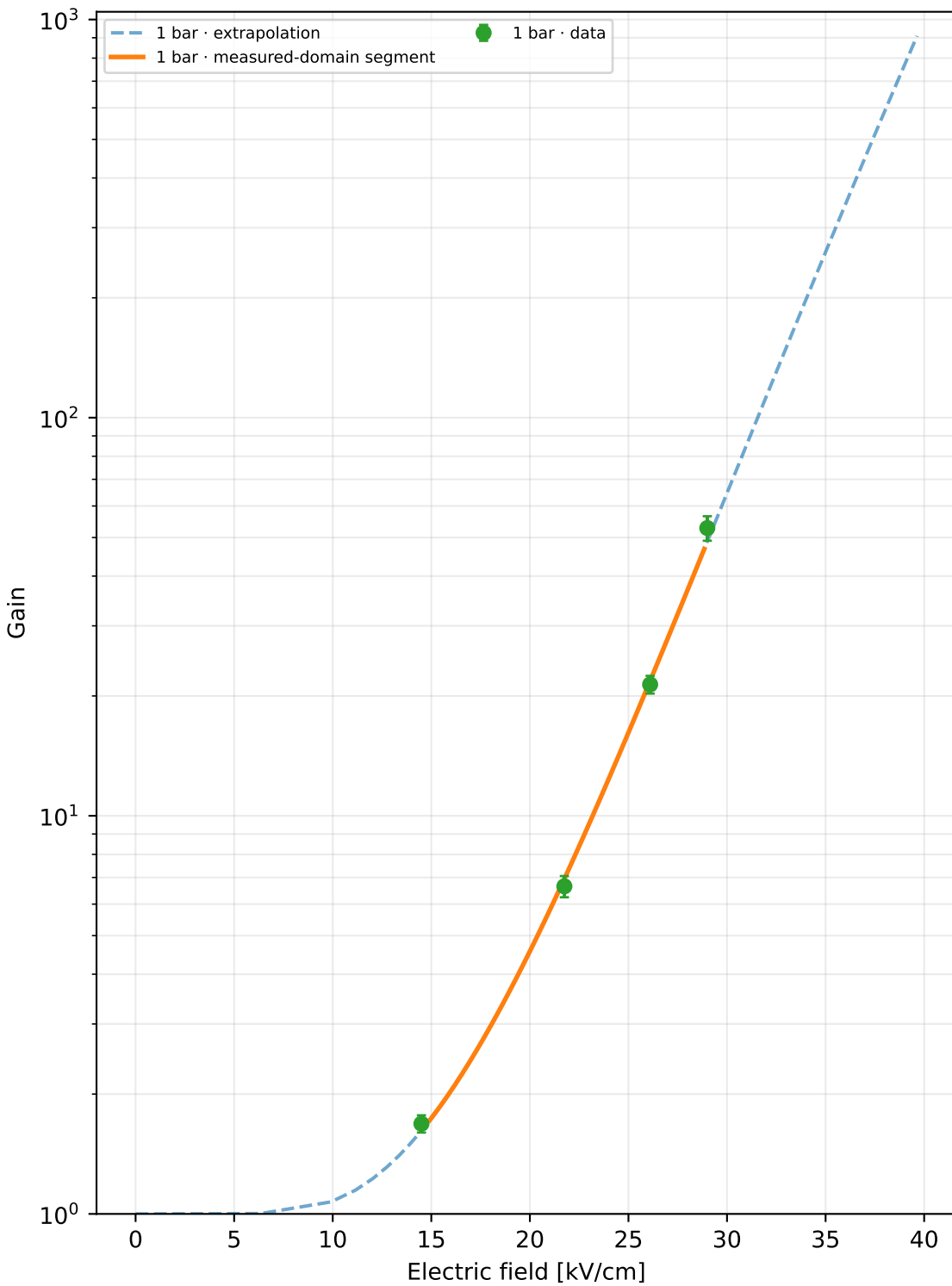
Gain-space residuals ($\pm \ln 1.2$ guides)



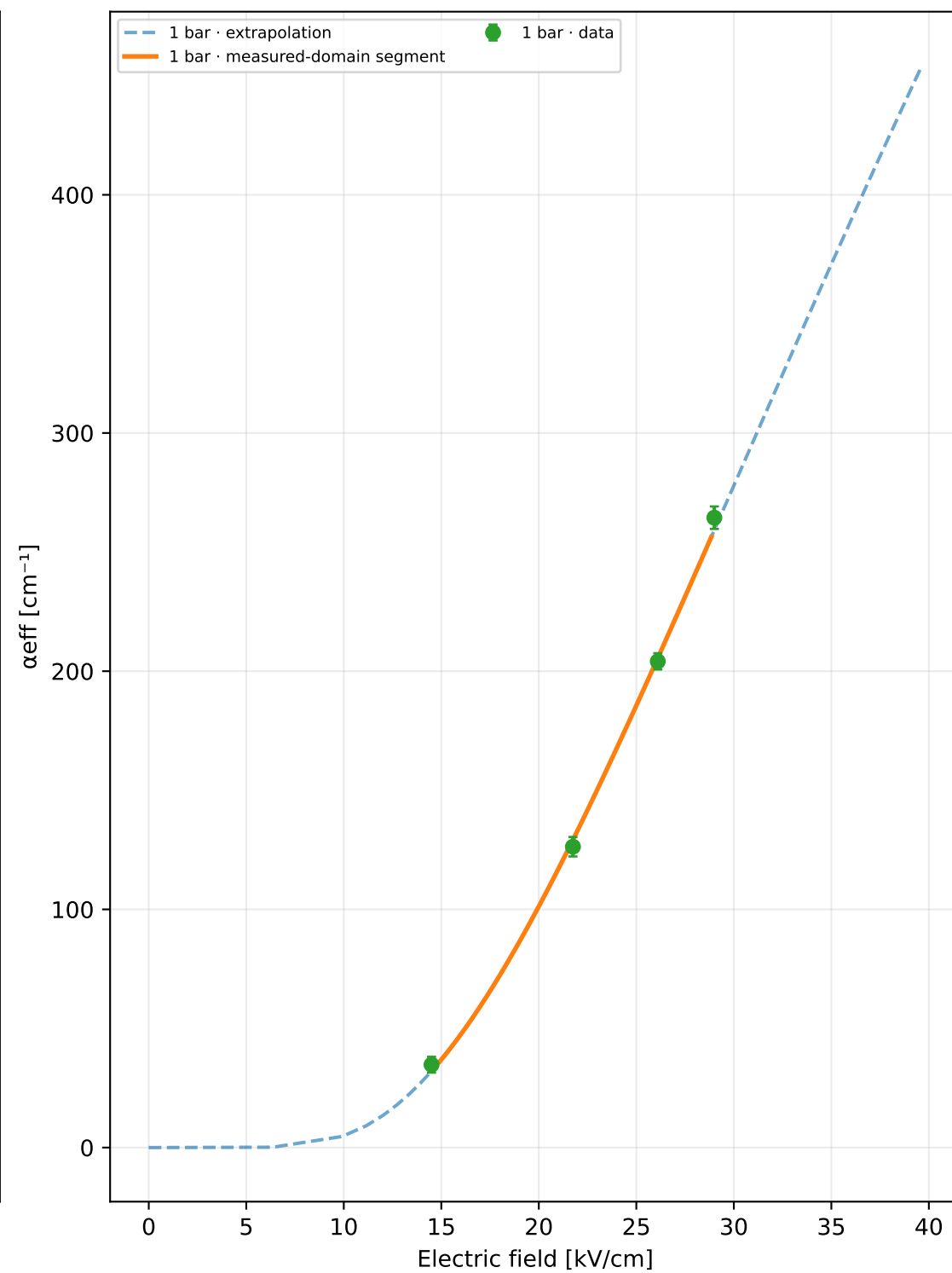
ROOT runs: 8
 Unique physical points: 4
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 1.044$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 14.5–29 $\text{kV cm}^{-1} \text{ bar}^{-1}$
 $A=2093.8$, $B=60.58$, $m=0$, $n=1$
 Covariance condition: 354
 Warnings: poor_cross_validation, large_cross_validation_outlier
 1 bar display limit: $G=909$ (local inversion limit)

Extrapolation capped at $G=1e+07$ or at the local model limit

Gain vs electric field



α_{eff} vs electric field



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	14.5	14.5	1.685	0.084	34.7844	200	2
1	21.75	21.75	6.65	0.408	126.308	200	2
1	26.1	26.1	21.37	1.1	204.133	200	2
1	29	29	52.81	3.74	264.447	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-9.16	—	—	no	poor cross validation, large cross validation outlier